The Impact of Noise Pollution on Standardized Test Performance

Josh Aarons*

4/16/26

Abstract

Noise pollution has become so ubiquitous in urban life that it is generally considered innocuous. However, research suggests that noise pollution can be more than a minor distraction and is associated with diminished health and productivity. I leverage exogenous shifts in noise pollution from a change in commercial flight paths and use a value-added model that accounts for student ability and school quality to estimate the causal impact of airplane noise pollution on standardized test performance. I find a large negative impact on math test performance, but no impact on English language art test performance. Moving from a school at the 25th percentile of exposure in my dataset to one at the 75th percentile is associated with a math test score reduction of 0.05 standard deviations, larger than the impact of missing 10 school days during the year. My findings suggest a potentially large value for exposed schools from investments in noise insulation.

*Department of Economics, University of California San Diego. Contact: jaarons@ucsd.edu

1 Introduction

High noise levels, even those well below the level required for hearing damage, are linked to poor sleep and adverse cardiovascular health outcomes (Münzel et al., 2014; Saucy et al., 2020; Vienneau et al., 2022), resulting in a significant public health burden (WHO, 2011). Even at relatively low levels, noise has negative non-health effects in the form of reduced productivity (Noweir, 1984; Dean, 2023). A particular concern with noise pollution is how pervasive it is. In the United States, more than 45 million people live within 300 feet of major transportation infrastructure such as a large road (EPA, 2014) and one in eleven public K-12 schools is within 500 feet (Hopkins, 2017).

Despite the pervasiveness of noise in modern life, there is only a nascent literature on the causal impacts of noise on cognition and learning. There are several reasons for the dearth of research including a lack of data, inability to disentangle air and noise pollution from traffic, and the perception that because noise is everywhere. Recently work has found that increased noise impairs memory and reduces productivity (Hygge, 2003; Jahncke et al., 2011; Dean, 2023). For the more than 4.4 million public K-12 school students attending schools within 500 feet of a freeway or major roadway, the traffic noise could distract students from learning and inhibit information recall on exams or reduce the effectiveness of teachers. The prevalence of noise could also diminish the defensive investments taken by schools. Prior research suggests workers are unaware that increased noise reduces their productivity and therefore do not make valuable defensive investments (Dean, 2023). In the case of K-12 education, schools may be under-investing in valuable noise insulation, exacerbating the negative impacts of noise pollution.

Motivated by the potentially large negative impact of noise pollution, cognitive psychologists have documented cross-sectional associations between noise and student performance (Stansfeld et al., 2005; Matheson et al., 2010) and comprehension (Haines et al., 2001; Stansfeld et al., 2005; Klatte, Bergström and Lachmann, 2013).[1] However, cross-sectional studies do not account for residential sorting. There is a well documented property value penalty associated with noise pollution (Crowley, 1973; Nelson, 1980, 2004). Therefore, students attending schools exposed to more noise pollution may be from lower income households than those attending schools in quieter areas. Given these results, researchers focused on educational outcomes should worry estimates relying on cross-sectional variation in noise

[1] There are a handful or experimental studies that find impacts of noise on memory tasks (Hygge, 2003; Hygge, Boman and Enmarker, 2003; Jahncke et al., 2011). These studies expose older, usually college-aged students to relatively low-level noise while reading and attempting to remember what they read. These studies unfortunately do not measure any impacts on standard measures of learning and do not attempt to determine whether noise impacts non-reading recall or skills. Potentially due to their often small sample sizes or participant motivation, the study results sometimes contradict each other (van Kempen et al., 2010).

capture both direct impacts from noise pollution on cognition as well as indirect impacts from residential sorting. In this paper, I present causal estimates of the impact of noise pollution on standardized test performance for public school students attending grades 3-8 near Seattle-Tacoma International Airport.

I leverage a change in commercial aircraft flight paths that shifted noise contours around Seattle-Tacoma International Airport (SeaTac) to use both across- and within-school variation in noise pollution over time. As part of the rollout of new technology, the Federal Aviation Administration (FAA) required airlines to install GPS transponders in all planes. GPS allows air traffic controllers to locate planes more precisely and permit them to fly closer together than was possible using older radar-based technology. The FAA used this advancement to shift more planes to the most efficient landing and takeoff routes to reduce approach and takeoff flight distance. This adjustment concentrated noise along certain corridors, resulting in a large increase in complaints from residents.[2] SeaTac implemented these new flight paths in 2015, resulting in an increase in the amount of noise under the most efficient flight paths. I utilize data from an array of noise monitors around SeaTac to measure the number of loud noise events from overhead airplanes near schools from 2009 through 2019 to capture this exogenous change to noise pollution.

I match noise pollution exposure to students' annual test scores to assess the impact of noise pollution on test scores. Specifically, I implement a value-added model of student performance in which prior year test scores, demographics, and school-grade characteristics all predict students' test scores. This approach adjusts for student ability to mitigate the impact of sorting on my estimates and generate a causal estimate of the effect of noise pollution. I estimate that additional loud flights near a school diminish student math test scores after accounting for other determinants of performance.

The noise pollution literature generally measures treatment as the total average background noise using the decibel scale. One issue in my context is that schools have vary significantly in their exposure to aircraft noise levels. Further, the decibel scale is exponential. As a result, there's no reason to think that a local average treatment effect is a good measure of the impact of changes in airplane noise. I therefore consider two nonlinear models of the share of school days at different levels of flight noise. My results suggest that the negative effects of noise pollution on test performance are driven by increases in the number of very loud days at the schools exposed to the most aircraft noise.

To the best of my knowledge, this is the first paper to estimate the causal impact of noise pollution on standardized test performance. My results show that noise pollution had a negative effect on math test performance but no impact on English language arts test

[2] *Advances in airport technology mean sleepless nights for some*

performance. This finding is consistent across an array of specifications. Under my preferred specification, I estimate that an additional loud flight overhead per hour during the school day (8 AM to 3 PM) diminishes math test scores by 0.0021 standard deviations. When considering the number of loud noise events per hour, this is a meaningfully large effect. Moving from a school in my dataset at the 25th percentile of exposure (2.2 loud flight events per hour) to the 75th (26.2 loud flight events) in 2019 is associated with a 0.05 SD reduction in math test scores, about the same impact as missing 12 days of school (Liu, Lee and Gershenson, 2021).

My work primarily contributes to the literature on identifying causal impacts of noise pollution in the labor context. Dean (2023) uses a randomized control trial at a textile training facility in Kenya to document a negative impact of noise on cognitive task productivity. Dean also elicits willingness to pay for quiet working conditions and finds that workers who are paid based on output are no more willing to pay for quiet conditions, suggesting workers are unaware that increased noise reduces their productivity. Hener (2022) shows that because wind direction determines the approach direction of planes at some airports, there is random variation in which direction planes approach the airport from on a given day. He therefore uses wind direction as an instrument for noise pollution in communities near Frankfurt Airport and finds that higher daily background noise increases the violent crime rate. I use a change in landing and takeoff paths to capture the effect of an unanticipated shift in exposure to aircraft noise pollution. I am not the first to use this policy change; Allroggen et al. (2025) and Nicolai V. Kuminoff (2025) leverage the same shift to estimate the effect of noise pollution on the Seattle, Chicago, and Boston and Phoenix housing markets, respectively. This change in flight paths impacted the distribution of noise pollution around many major airports in the United States, which could provide contexts for future research on noise pollution.

While the noise pollution literature is still sparse, there is a robust literature measuring the causal impacts of pollution on education and human capital accumulation. It is well documented that contemporaneous air pollution reduces exam scores (Ebenstein, Lavy and Roth, 2016; Graff Zivin et al., 2020; Zhang, Chen and Zhang, 2018). I focus on cumulative exposure over the course of the school year, which has also been studied in the air pollution context (Heissel, Persico and Simon, 2022; Wen and Burke, 2022; Stafford, 2015). Beyond diminishing test scores, air pollution is tied to increased absence rates and behavioral incidents (Heissel, Persico and Simon, 2022). I extend the work on the adverse impacts of pollution exposure on education to include noise pollution.

The remainder of this paper is organized as follows. Section 1 provides information on noise levels and the implementation of new flight paths at SeaTac. Section 3 explains how the

noise data is collected and merged with school grade-level test scores. Section 4 explains the value-added model and strategy for estimating the impact of loud flights on test performance. Section 5 presents the results from the value-added model. Section 6 provides nonlinear estimates of noise pollution impacts aimed at understanding the relationship between noise and student performance. Finally, Section 7 concludes.

2 Context

2.1 Performance-Based Navigation

Over the past two decades, the Federal Aviation Administration (FAA) has made a series of changes to commercial flight paths and procedures. The backbone of this suite of programs commonly known as NextGen is an advancement in navigation technology. Starting in the early 2010s, the FAA began using Automatic Dependent Surveillance-Broadcast (ADS-B) technology to track planes and mandated all planes be equipped with ADS-B by 2020. ADS-B determines an airplane's position using GPS (satellites) rather than radar (radio waves), which allows for more precise location tracking.

Using ADS-B, the FAA implemented Performance-Based Navigation (PBN), a program that altered the routes pilots used for takeoff and landing, at the nation's busiest airports. The program sets limits on flight paths, giving air traffic controllers and pilots less autonomy in determining routes. With more precise location information from ADS-B, air traffic controllers can safely have planes fly in closer proximity rather than encouraging pilots to spread out over a wider area. This has concentrated more planes on the most efficient routes for takeoff and landing.

PBN has caused noise corridors to tighten and become louder as planes fly over the same optimal paths in quick succession, leading to pushback from members of the public. Airports and the FAA have seen dramatic increases in noise complaints,[3] which seems to be due to an increase in the number of loud noise events since the distribution of noise levels is similar before and after the implementation (see Appendix Figure 1).[4] When flight paths shift, there are both winners and losers in terms of noise pollution – some areas will see declines and other increases. This project does not attempt to measure the impact on total social surplus, but to instead estimate the effect on a subset of those who were affected by the path changes.

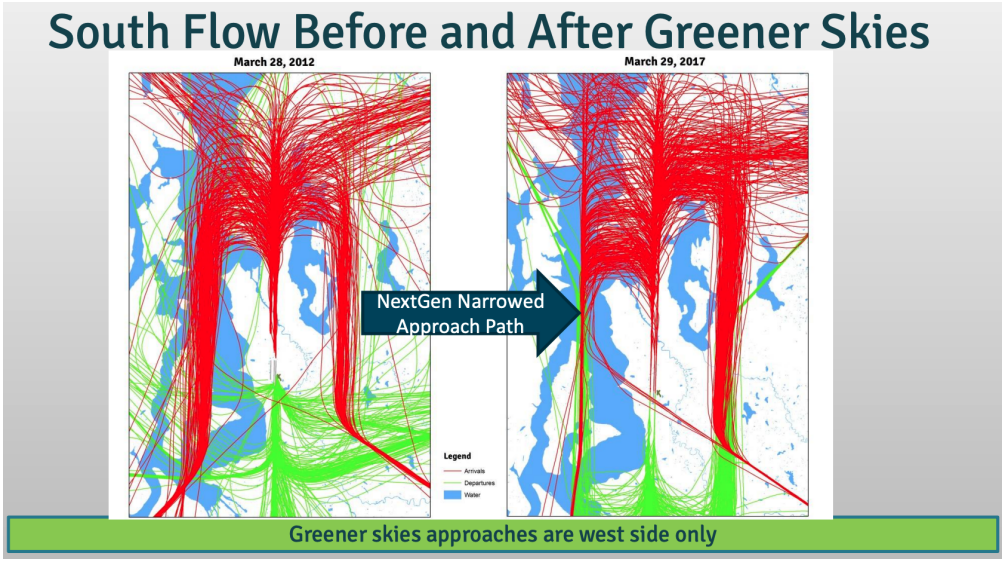
[3] *Advances in airport technology mean sleepless nights for some* and *New flight paths lead to airplane noise complaints across US*

[4] These complaints do not impact the way that air traffic controllers route flights day to day, though a lawsuit in Phoenix did lead to temporary reversion to the previous flight paths.

2.2 Seattle-Tacoma Airport and Noise Pollution

The FAA piloted new PBN flight paths on 1% of flights at SeaTac beginning in March 2013 (United States Department of Transportation, 2014). SeaTac expanded the program to all planes equipped with ADS-B technology in April 2015 (Alaska Airlines, 2015). As a result, flight paths used for landings and departures concentrated along more narrow corridors, which is particularly noticeable for southern departures depicted in green in Figure 1.

Figure 1: Change in Air Traffic Flow for Southern Departures and Northern Arrivals



SeaTac also experienced a steady increase in flight traffic during the 2010s. In particular, the switch to PBN in 2015 allowed for greater annual growth in air traffic. As a result, PBN induced both an immediate shock and gradual increase in aircraft noise pollution (see Appendix Figure 2).

3 Data and Treatment Measurement

3.1 Loud Noise Events

The Port of Seattle operates a network of noise monitors around SeaTac, and 24 of their monitors covered 2009 through 2020 (Figure 2). These monitors record sound pressure continuously and document all time spans that correspond to an airplane flying overhead. The Port of Seattle quantifies the sound pressure from flights with a commonly used measure, sound exposure level (SEL). My data capture each time the SEL exceeds 65 decibels (dB), approximately the same volume as using a vacuum cleaner or standing next to city traffic.

The vast majority of the events captured by the monitors are not loud enough to cause hearing damage unless they were to continue unabated for multiple hours, but they are loud enough to alarm, distract, or drown out talking.

I measure treatment as the number of SEL events per hour from 8 AM to 3 PM on school days.[5] The school year runs from the start of September to the start of June in Washington, and schools are allowed to administer standardized tests any time during the final 12 weeks of the school year. I do not have data on the date of exams and therefore cannot determine the amount of exposure during exams. I calculate the number of SEL events per school hour from September through the end of February to focus on the effect of cumulative exposure during learning. However, noise pollution during March, April, and May is highly correlated with noise pollution during the rest of the school year, so I cannot separate the impact of noise pollution during exams on recall from the impact of cumulative noise pollution on learning.

One difficulty with the SEL data is that 26% of school hours in my sample have no SEL events. This could either mean that there were no loud flights overhead during that hour or that the monitor was not collecting data. The probability of not recording any SEL events is non-random both across monitors and days. For example, the share of monitor-days with no SEL events is lowest in December when more people are flying. To address occasions where data is missing across all stations, I drop all days where at least three quarters of stations do not record a single SEL event.[6] These days are concentrated in five school year months (November and December 2010 and January, February, and November 2015) which are missing data. Of the 115 September - February school days I classify as missing data, 108 are during these five months. Unfortunately, most monitors are missing all data for the first four months of 2015. Without data for the start of the year and April, when flight paths changed, I cannot tell whether flight volumes changed notably between the learning period and exam period in 2015. I choose to drop the 2014/15 school year from my data because if there is a large increase during the exam period in 2015, then including the school year would increase measurement error. I conduct sensitivity analyses including the 2015 tests and show that this choice does not change my results.

To address cases where specific monitors are not collecting data, I count consecutive days with no SEL events. I drop all hours during spells of at least two weeks without an SEL

[5] I choose to focus on the school day as these are the hours I know students are exposed to the measured noise. There is no shock to hour to hour variation in SEL events, so I cannot disentangle the effects of school day versus evening or weekend noise (and I lack student residence locations).

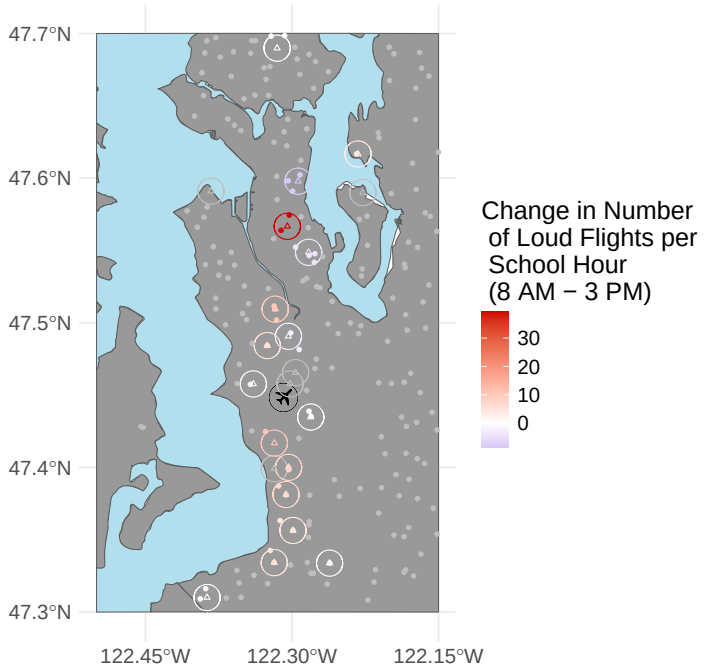
[6] This is to remove the 5% of days with a high missing rate, which are usually days where all monitors or around 80% of monitors are missing data. Only 1.2% of days have missing rates between 50% and 75%, so the decision to set the threshold at 75% is not impactful.

event. This process removes 3.6% remaining monitor-hours. The two-week threshold trades off underestimating the number of SEL events by including hours when monitors aren't functioning with overestimating by excluding hours where there were no flights overhead for many days in a row. Changing the threshold by a week in either direction has a very small impact on the share of hours dropped and no impact on the results.

3.2 Student Data

Washington state's Education Research and Data Center provided data on student enrollment, demographics, and test scores for this project. My outcomes of interest are annual performance on mathematics and English language arts (ELA) standardized tests taken by all third through eighth grade students. I measure a student's performance on a test as number of standard deviations from the statewide mean that year. This accounts for year-to-year changes to the testing instrument and difficulty.

Figure 2: Monitor and School Matching



The Washington Office of Superintendent of Public Instruction (OSPI) provides data on school locations. I match monitors to all schools within one kilometer.[7] Figure 2 provides a visual representation of this process. Each monitor is depicted as a triangle inside a circle, where the circle measures the one kilometer buffer around each monitor. The color

[7]Later, I show that my results are not sensitive to this specification.

of each monitor shows the change in average annual loud flights overhead induced by the change in flight paths (specifically the difference between the 2015-2019 and 2009-2014 school year averages). Monitors that experienced the largest increases in SEL events are the most red, while white monitors saw little change and pale blue monitors experienced less traffic. Schools inside the buffer around a monitor are matched and depicted using the monitor's color rather than the gray of the unmatched schools. My final dataset includes school years for all third through eighth grade public school students who were enrolled at a single school during the school year.

4 Empirical Design

4.1 Value-Added Model

In my setting, treatment is at the school level and there is no variation in treatment exposure within schools during a school year. Variation in noise pollution comes from assignment to schools, year to year increases in number of flights in and out of SeaTac, and the shift in flight paths. Additionally, since all students in a grade take the same exams, both the treatment and the testing instrument that I use to measure outcomes is the same across all students in grade g at school s. I choose to measure outcomes at the student level rather than school-grade level to account for changes in student composition that could correlate with treatment.

I identify the impact of noise pollution on student performance by implementing a value-added model. The logic of the value-added model is that students perform similarly across years, so past test scores are a good estimator of future test scores. I model student i's standardized test performance as a function of the number of loud noise events at the student's school s during school year t, a vector of student characteristics, X_{it}, that includes student demographics and their test performance from the prior year, and idiosyncratic student-level variation, ε_{it}. To account for variation in testing instrument, I measure performance, y_{it}, as the number of standard deviations from the mean score for all Washington students in that grade g during year t. Since I do not have teacher or classroom identifiers, I include school-grade fixed effects. I estimate the following equation using ordinary least-squares (OLS) regression:

$$y_{it} = \beta \boldsymbol{X}_{it} + \gamma SEL_{st} + \rho PM_{st} + \lambda_{sg} + \varepsilon_{it} \tag{1}$$

SEL_{it} measures the amount of noise pollution the student experiences that year. The pa-

rameter of interest is γ, which is the estimate of the causal impact of an additional loud noise event per hour during a school year. The vector $\boldsymbol{X}_{it}$ includes lagged test performance, gender, race/ethnicity, a flag for whether English is not their primary language spoken at home, and whether they receive free or reduced-price lunch during the school year. PM_{st} is the average daily $PM_{2.5}$ the school was exposed to during the school year.[8] λ_{sg} are time-invariant school-grade fixed effects that account for any differences in school composition not captured by demographics. These could include teacher quality or unobserved differences in average students for example.

4.2 Smarter Balanced Assessments Implementation

One complication with my context is that Washington switched from testing students using the Measurements of Students Progress (MSP) to the Smarter Balanced Assessment system (SBA) in 2014/15. This means that a large shift in noise pollution coincided with a shift in test instrument. The left panel of Figure 3 shows that students were less likely to meet the exam standard after the switch to the SBA than they were before.

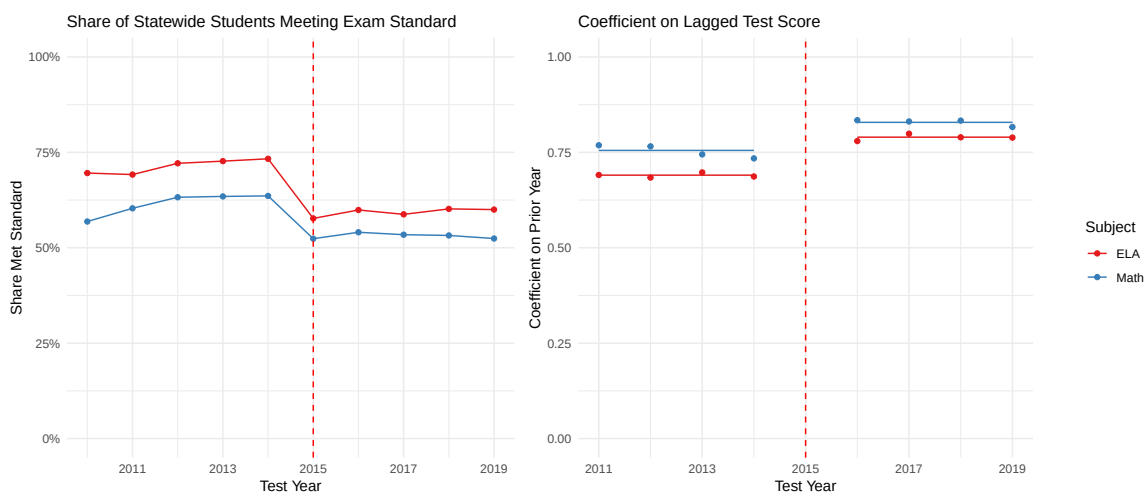


Figure 3: Impact of Test Regime Change

The regime change creates four problems for this analysis. First, students are less likely to pass the new exam and the tests have different scoring scales. I use standard deviations from the state mean to scale everyone's performance so that this impact does not affect the results.[9] Second, prior to 2014/15, all third through eighth grade students took a reading

[8] After adjusting for the school and year averages, the number of SEL events is uncorrelated with the average daily $PM_{2.5}$.

[9] One exception would be if the change in test instrument had differential impacts across students of different demographics. I show in Appendix Figure 4 that there do not appear to be notable differences in impacts across demographic groups.

exam and both fourth and seventh grade students also took a writing exam. The SBA assessments used from 2015 onwards are composed of just a single ELA exam administered to all students. To minimize the impact of this change, I use the reading exam only prior to 2015 and the ELA exam from 2015 onward. Third, during the final year of the MSP regime, schools could choose whether or not to administer the old exam, resulting in a smaller sample that year. The value-added model I use addresses concerns about selection into test taking by conditioning on prior year performance, however I also conduct sensitivity analysis around the inclusion of this year and find no impact on my results.

The final concern pertains to the prior year test score variable in the value-added model. Lagged test scores serve as a measure of student ability, but if their predictive value varies from year to year, then residuals will be correlated with year. This is particularly concerning since the testing regime and flight paths change during the same year. The right panel of Figure 3 shows that the SBA exams better predict subsequent exam performance than the MSP exams do. In this figure, each point is the coefficient on the previous year's exam performance from a regression of performance on prior year performance interacted with test year. The horizontal lines are the coefficients from a separate regression where prior year performance is interacted with assessment regime instead of year. The annual and assessment regime coefficients show that each year's testing instrument is almost exactly as predictive as other instruments used under that assessment regime. In my model, I interact prior year performance with the assessment regime to account for the difference in testing instruments. This solves the problem of using a single variable for lagged performance, which would overestimate the prediction value of the prior year performance variable under the MSP regime and underestimate under the SBA regime, potentially leading to more residual variation to explain during the years that noise pollution levels grow and diverge.

5 Results

I present my primary results from estimating the value-added model using OLS in Table 1. The coefficient on loud noise events represents the average impact of an additional flight per hour on standardized test performance measured in standard deviations (SDs) above or below the annual mean test score. My estimates suggest a negative impact of additional loud noise events on math test scores, but not on ELA test scores.

I estimate that an additional loud noise event per school hour reduces standardized math test scores by 0.0021 SDs. The magnitude of these estimates is fairly large. Heissel, Persico and Simon (2022) estimate that attending a school just downwind of a highway instead of one just upwind of a highway reduces standardized test scores by 0.04 SDs due to the large

Table 1: Impact of One Additional Loud Noise Event per Hour on Exam Performance

	Math	ELA
Loud noise events per hour	-0.0021***	0.0002
	(0.0005)	(0.0004)
Observations	65,855	65,838
R^2	0.740	0.679

*/**/*** denote significance at the $p < 0.1/0.05/0.01$ level. Test performance is measured in standard deviations from the mean score. Standard errors are clustered at the school level.

increase in exposure to air pollution.[10] Moving from a school in my dataset at the 25th percentile of exposure during the 2019 school year (2.2 loud flights per hour) to the 75th (26.2 loud flights per hour) is associated with a 0.05 SD reduction in score, about 20% larger than the impact of missing 10 school days during the year (Liu, Lee and Gershenson, 2021).[11]

5.1 Sensitivity Analysis

I conduct several sensitivity analyses to determine whether my findings are robust to alternate specifications. First, I consider whether my choices of matching radius around noise monitors or minimum decibel level are driving my results. Expanding the radius around noise monitors increases the sample size, but reduces the accuracy of the noise pollution variable. Each additional matched school is further from the noise monitor, likely inducing more measurement error. Figure 4 shows that for all buffers considered from 500 to 2000 meters the estimate for the impact of noise pollution on math test scores is significant, but it becomes highly imprecise when the buffer shrinks to 250 meters.

I also consider whether my choice to exclude the 2015 test year and include the 2014 test year drives my results. I excluded the 2015 test year from my preferred specification as the flight path change happens in April of 2015 and I am missing data for January through April 2015. I run a regression including the 2015 tests where the measure of noise pollution is the average number of SEL events per school day during September through December 2014 for the observations that year. I also consider excluding the 2014 test year as schools

[10] Based on their first stage, the effect of an additional microgram per cubic meter of $PM_{2.5}$ is 0.02 SDs, which is similar to the 0.016 that Gilraine and Zheng (2024) find.

[11] Goodman (2014) uses moderate snowfall as an instrument for absences and estimates a much larger impact of absences on math test scores (0.05 SDs per absence) than do Liu, Lee and Gershenson (2021). However, he notes that if teacher absences increase during moderate snowfall days, then this estimate captures the effect of teacher absences as well.

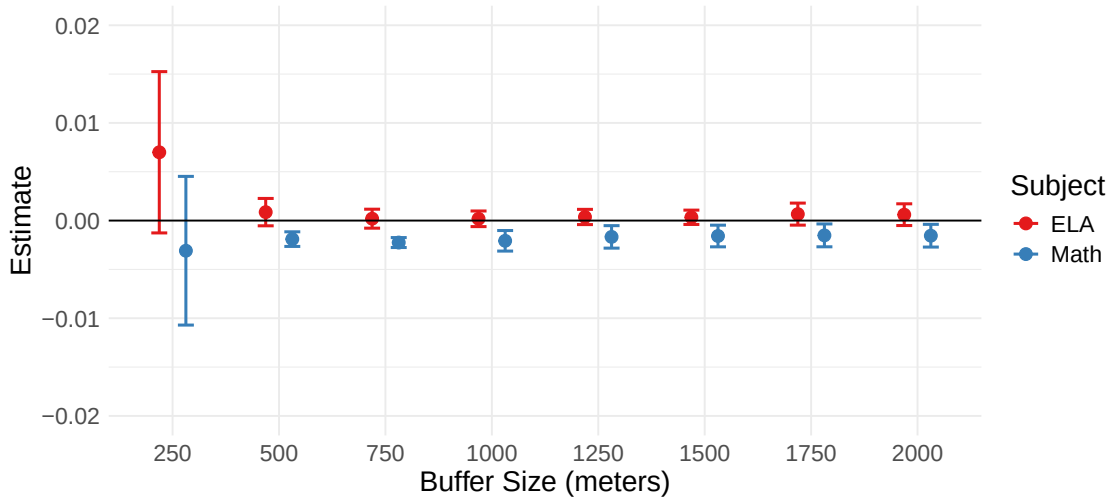


Figure 4: Robustness to Alternate Radius Distance Choices

had flexibility in choosing whether or not to administer the exam that year. I compare my primary specification to these two specifications in Table 2. The results are similar across all specifications.

Table 2: Robustness to Alternate Year Specifications

	Math			ELA		
	(1)	(2)	(3)	(4)	(5)	(6)
SEL events per hour	-0.0021***	-0.0020***	-0.0020***	0.0002	-0.0004	0.0005
	(0.0005)	(0.0004)	(0.0005)	(0.0004)	(0.0004)	(0.0004)
Includes 2014	X	X		X	X	
Includes 2015		X			X	
Observations	65,855	71,001	60,840	65,838	70,957	60,822
R^2	0.740	0.738	0.745	0.679	0.676	0.686

*/**/*** denote significance at the $p < 0.1/0.05/0.01$ level.
Preferred model excludes 2015 test year due to change in flight paths in April 2015.

6 Nonlinear Noise Impacts

The previous section shows evidence that more loud noise events from overhead flights negatively impact performance on standardized math tests, but it does not attempt to quantify the impact of additional noise. Much of the literature on adverse impacts of noise

pollution (Dean, 2023; Hener, 2022; Nicolai V. Kuminoff, 2025; von Graevenitz, 2018) uses equivalent continuous sound level (Leq), a measure of average noise over a period of time, as the measure of noise pollution. The Port of Seattle also collects Leq data and calculates a level for both community and aircraft noise. In the remainder of this section, I use aircraft noise Leq to understand the impact of the amount of aircraft noise rather than the number of flights.

6.1 Noise Levels

The intensity of sound is measured using the decibel scale. The decibel scale is a logarithmic scale, where a 10 decibel (dB) increase indicates 10 times as much sound energy. However, the human ear only perceives a doubling in volume, so humans are likely to underestimate the intensity (in terms of sound pressure) of a noise increase. According to the National Institute for Occupational Safety and Health, noises louder than 85 dB can cause hearing damage from sustained exposure (or sufficiently loud noises).[12] Only about a quarter of recorded noise events in my sample are above 85 dB, while the majority of noise exposure is to lower level noises about as loud as a vacuum cleaner or coffee grinder. Appendix Figure 3 provides the distribution of observed sound exposure level events with additional examples to elucidate the noises students are exposed to in this context.

Leq is calculated by taking the logarithmic average of the total sound pressure over a period. An important feature of Leq is that the loudest noises determine the Leq. For example, consider one hour at two different noise monitors: at one there was one flight overhead with a SEL of 90 dB, at the other there were 100 flights overhead each with SELs of 70 dB. The decibel scale is logarithmic, so the 90 dB flight generated 100 times as much sound pressure as each of the 70 dB flights. Therefore, the Leq of these two monitor-hours is the same even though one had nearly two loud flights per minute.[13] This example shows how the loudest noises during an hour, such as those from airplanes, construction, or cars without mufflers, will be the primary determinant of Leq levels.

[12] National Institute for Occupational Safety and Health: https://www.cdc.gov/niosh/noise/about/noise.html

[13] My data compress each loud noise event into SEL, which is a measure of total sound pressure compressed into one second. With the complete recording data, one could calculate Leq a different way that better accounts for the peak and duration of each event. In my case, aircraft flyovers should fairly similar in duration, and this method should not mask notable differences in max volume and duration.

6.2 Empirical Model

I measure average noise levels during a school year as the average hourly Leq during all school hours during the school year. The Port of Seattle splits Leq into aircraft noise (based on noise levels while airplanes are flying overheard) and community noise (all other noise). I calculate the average school day aircraft Leq by aggregating hourly total flight noise over school hours and taking the logarithmic average. Figure 5 shows the distribution of daily school day aircraft Leq across noise monitors.

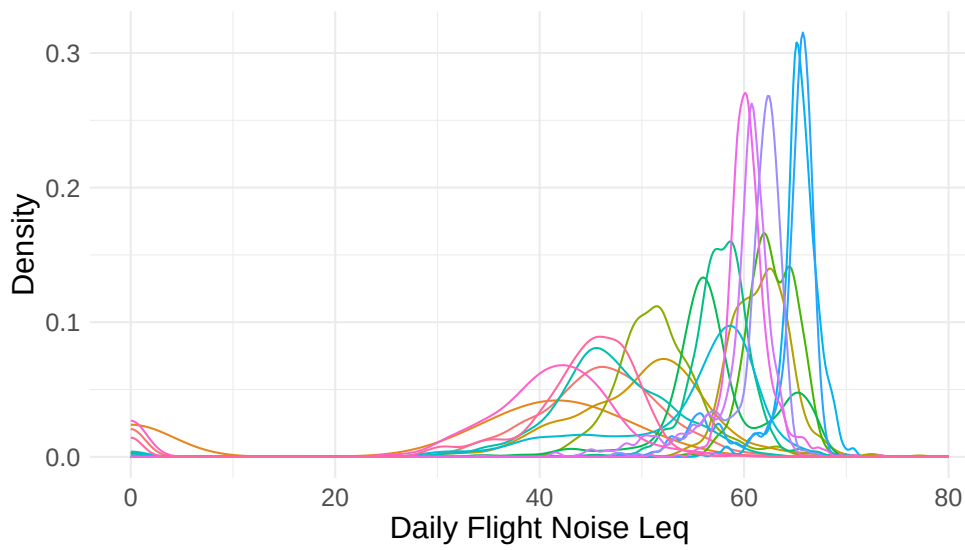


Figure 5: Distribution of School Day Aircraft Leq by Noise Monitor

I set up a nonlinear model of the impact of the share of days at different levels of noise exposure. This takes the form:

$$y_{it} = \beta \boldsymbol{X}_{it} + \sum_{j \in J} \gamma_j FlightLeq_{mt} + \rho PM_{st} + \lambda_{sg} + \varepsilon_{it} \tag{2}$$

where $FlightLeq_{mt}$ is the share of days in each decibel range bin. Ideally, I would create bins with relatively small Leq ranges to count the number of days at different aircraft noise levels and estimate the effect of having an additional school day with a specific aircraft Leq instead of a reference Leq. Under the belief that the amount of noise impacts student performance rather than the amount relative to the school's average daily noise, this would provide estimates of the effect of exposure to different amounts of noise. One complication with this approach is that the support of the aircraft Leq distribution varies across noise monitors. Therefore, it is not possible to estimate the causal effect of switching from a 45

dB aircraft noise day to a 70 dB day.

I handle the lack of overlap of support by setting a relatively large reference bin to compare other days against so that all monitors have days in the reference bin. My primary specification has nine decile bins (<42 dB, 42-46, 46-50, 50-60, 60-62, 62-64 64-66, 66-68, $\geq$ 68). I also estimate a model that instead uses five bins, one for each quintile of aircraft Leq. It is worth noting that estimates of impacts of additional exposure at the upper end of the distribution come from comparing greater exposure years at louder schools to lower exposure years at those schools while estimates of the impacts of additional exposure at the lower end of the distribution come from comparing lower exposure years at lower exposure schools to higher exposure years at those schools.

6.3 Results

My nonlinear model[14] does not yield conclusive evidence. Figure 6 shows that the point estimates for the effect of more days at higher flight noise Leq bins are negative with increasing magnitude in decibel level, but generally not significant at the $p < 0.05$ level. As a result, I cannot rule out a large effect of the noisiest days. Moreover, all estimates for days with flight noise greater than 62 dB are negative, and 30% of schools in the sample had at least $\frac{2}{3}$ of school days with flight noise at least 62 dB, suggesting the potential for a large amount of impact on schools. My quintile regression reiterates this result (see Appendix Figure 7). An additional 10% of school days with flight noise in the highest quintile, which corresponds to days with flight noise of 62.7 dB or greater, is associated with a significant 0.163 SD decrease in math standardized test scores, a large negative effect.

The lack of coverage of exposure to days with different amounts of flight noise limits both the precision and interpretation of my model. There are more than 180 schools within 20 kM of SeaTac, but I can only match one in six to a noise monitor, meaning many treatments and responses remain unmeasured, inhibiting my ability to estimate a dose response curve. Estimates of noise from flight trajectory data, such as those used by Allroggen et al. (2025), could allow for more coverage of student treatment levels and prove more fruitful than monitor data for nonparametric estimates, such as those necessary fo estimate a dose response curve.

[14] I avoid calling this model nonparametric as the large bin sizes place assumptions on the equal impact of days with different amounts of flight noise.

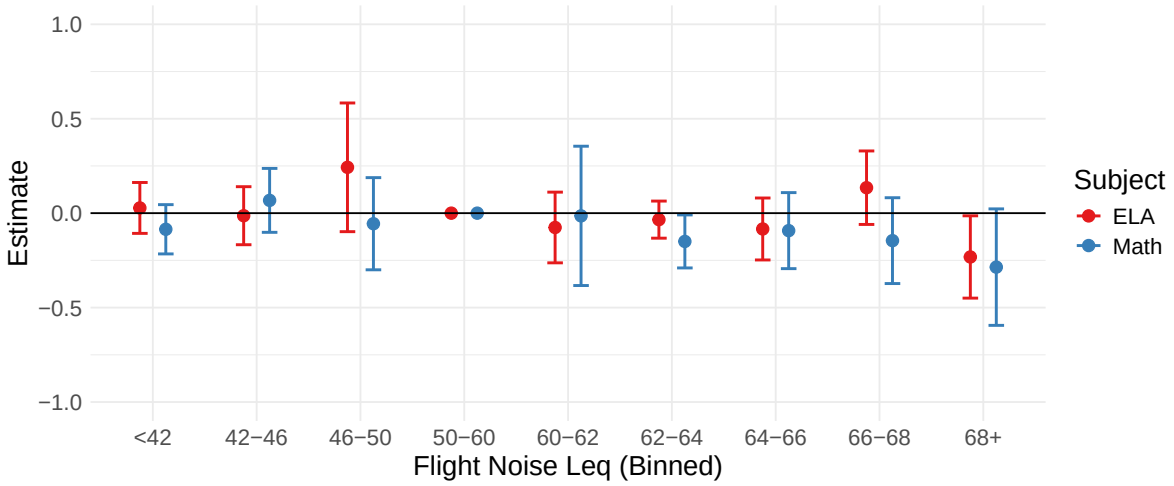


Figure 6: Estimate of Increased Share of Days in Flight Noise Leq Bin on Annual Standardized Test Performance

7 Conclusion

This is the first study, to my knowledge, to provide causal estimates of the impact of noise pollution on standardized test performance. I use an exogenous shift in aircraft noise to assess the impact of noise pollution on student performance. I find a meaningfully large impact of loud noises during the school day on math test scores but not on ELA test scores. Further, my estimates suggest that the magnitude of these results is large. Moving from a school in my dataset at the 25th percentile of exposure (2.2 loud flight events per hour) to the 75th (26.2 loud flight events) in 2019 is associated with a 0.05 SD reduction in math test scores, about the same impact as missing 12 days of school (Liu, Lee and Gershenson, 2021).

A limitation of my work is that I cannot disentangle what share of the estimated impact is from cumulative impacts on reduced learning and what share is from contemporaneous impacts on exam days. Washington allows schools to set their own testing dates and times (within a range of a couple months), and the school districts I contacted have not retained this information for the period of my study. As a result, I cannot determine the amount of pollution during test times and would need to use the average amount during the testing period instead. Average noise during the testing period is highly correlated with noise during the rest of the school year. Therefore, I cannot disentangle the effects on learning and encoding information from the effects on test taking and information recall. Future research should address this issue by leveraging a setting where there is variation between cumulative noise pollution exposure and exposure during the test. One possibility would be to take the approach of Ebenstein, Lavy and Roth (2016) and measure day-of-test exposure controlling for prior exposure in a setting where there is sufficient daily variation. Many

airports change flight paths depending on the direction of the wind (Hener, 2022), so this should be possible in an airplane noise pollution context.

My finding of a negative impact of noise pollution on academic performance suggests two immediate steps for reducing the impacts of noise pollution. First, in the school context, sound insulation may mitigate damages. SeaTac and the Port of Seattle have been proactive about reducing noise pollution – their noise pollution monitoring system is more thorough and has been around longer than other major airports, and they've focused on methods to reduce noise pollution. Additionally, the Port of Seattle and FAA have spent hundreds of millions of dollars funding sound insulation projects for houses to mitigate pollution, and recently allocated $70 million to insulate nine schools (with $30 million set aside for six more).[15] This funding was focused on the Highline School District, which is closer to the airport than many of the schools treated with an increase in loud noise events following the PBN implementation. This suggests a role for expanding the funding to schools further from the airport, but underneath flight paths. Future research should investigate noise insulation as a mitigating factor on the impact of noise pollution on cognition and and learning.

In line with the previous recommendation, most airports do not currently have a robust noise monitoring system. The areas identified as affected by airport noise are generally very close to airports, so sound proofing and other assistance is focused on these neighborhoods. My findings suggest large impacts to schools further away (a cluster of schools that were impacted most negatively by the flight path changes was 8-10 miles – 13-16 kilometers – away from the airport). Ensuring adequate data collection to measure impacts to communities further from the airport would allow for monitoring of the impacts of noise pollution. Further, if airports or the FAA see benefits from adjusting flight paths, they should consider offsetting or mitigating damages to negatively affected communities by funding insulation or another form of transfer payment.

[15]The FAA funds insulation for houses exposed to excessive noise pollution from air travel, and the Port of Seattle has used airport revenues to fund additional sound insulation as well: https://www.westsideseattle.com/highline-times/2020/02/26/port-commission-approves-acceleration-noise-insulation-efforts-benefit.

References

Alaska Airlines. 2015. "Report Shows Benefits of Greener Skies Approaches at SeaTac Airport." https://news.alaskaair.com/newsroom/report-shows-benefits-of-greener-skies-approaches-at-sea-tac-airport/.

Allroggen, Florian, R. John Hansman, Christopher R Knittel, Jing Li, Xibo Wan, and Juju Wang. 2025. "Planes Overhead: How Airplane Noise Impacts Home Values." National Bureau of Economic Research Working Paper 34431.

Crowley, Ronald W. 1973. "A Case Study of the Effects of an Airport on Land Values." *Journal of Transport Economics and Policy*, 7(2): 144–152.

Dean, Joshua. 2023. "Noise, Cognitive Function, and Worker Productivity." *American Economic Journal: Applied Economics, forthcoming.*

Ebenstein, Avraham, Victor Lavy, and Sefi Roth. 2016. "The Long-Run Economic Consequences of High-Stakes Examinations: Evidence from Transitory Variation in Pollution." *American Economic Journal: Applied Economics*, 8(4): 36–65.

EPA. 2014. "Near Roadway Air Pollution and Health: Frequently Asked Questions." Environmental Protection Agency.

Gilraine, Michael, and Angela Zheng. 2024. "JUE insight: Air pollution and student performance in the U.S." *Journal of Urban Economics*, 143: 103686.

Goodman, Joshua. 2014. "Flaking Out: Student Absences and Snow Days as Disruptions of Instructional Time." National Bureau of Economic Research Working Paper 20221.

Graff Zivin, Joshua, Tong Liu, Yingquan Song, Qu Tang, and Peng Zhang. 2020. "The unintended impacts of agricultural fires: Human capital in China." *Journal of Development Economics*, 147(C): S0304387820301358.

Haines, MM, SA Stansfeld, RF Job, B Berglund, and J Head. 2001. "Chronic aircraft noise exposure, stress responses, mental health and cognitive performance in school children." *Psychological Medicine*, 31: 265–277.

Heissel, Jennifer A., Claudia Persico, and David Simon. 2022. "Does Pollution Drive Achievement? The Effect of Traffic Pollution on Academic Performance." *Journal of Human Resources*, 57(3): 747–776.

Hener, Timo. 2022. "Noise pollution and violent crime." *Journal of Public Economics*, 215: 104748.

Hopkins, Jamie Smith. 2017. "The invisible hazard afflicting thousands of schools." the Center for Public Integrity.

Hygge, Staffan. 2003. "Classroom experiments on the effects of different noise sources and sound levels on long-term recall and recognition in children." *Applied Cognitive Psychology*, 17(8): 895–914.

Hygge, Staffan, Eva Boman, and Ingela Enmarker. 2003. "The effects of road traffic noise and meaningful irrelevant speech on different memory systems." *Scandinavian Journal of Psychology*, 44(1): 13–21.

Jahncke, Helena, Staffan Hygge, Niklas Halin, Anne Marie Green, and Kenth Dimberg. 2011. "Open-plan office noise: Cognitive performance and restoration." *Journal of Environmental Psychology*, 31(4): 373–382.

Klatte, Maria, Kirstin Bergström, and Thomas Lachmann. 2013. "Does noise affect learning? A short review on noise effects on cognitive performance in children." *Frontiers in Psychology*, 4: 578.

Liu, Jing, Monica Lee, and Seth Gershenson. 2021. "The short- and long-run impacts of secondary school absences." *Journal of Public Economics*, 199: 104441.

Matheson, Mark, Charlotte Clark, Rocio Martin, Elice van Kempen, Mary Haines, Isabel Lopez Barrio, Staffan Hygge, and Stephen Stansfeld. 2010. "The effects of road traffic and aircraft noise exposure on children's episodic memory: the RANCH project." *Noise Health*, 12: 244–254.

Münzel, Thomas, Tommaso Gori, Wolfgang Babisch, and Mathias Basner. 2014. "Cardiovascular effects of environmental noise exposure." *European Heart Journal*, 35(13): 829–836.

Nelson, Jon P. 1980. "Airports and Property Values: A Survey of Recent Evidence." *Journal of Transport Economics and Policy*, 14(1): 37–52.

Nelson, Jon P. 2004. "Meta-Analysis of Airport Noise and Hedonic Property Values: Problems and Prospects." *Journal of Transport Economics and Policy*, 38(1): 1–27.

Nicolai V. Kuminoff, Jasdeep Mandia. 2025. "Valuing Noise Pollution: Evidence from Flight Path Changes."

Noweir, Madbuli H. 1984. "Noise exposure as related to productivity, disciplinary actions, absenteeism, and accidents among textile workers." *Journal of Safety Research*, 15(4): 163–174.

Saucy, Apolline, Beat Schäffer, Louise Tangermann, Danielle Vienneau, Jean-Marc Wunderli, and Martin Röösli. 2020. "Does night-time aircraft noise trigger mortality? A case-crossover study on 24,886 cardiovascular deaths." *European Heart Journal*, 42(8): 835–843.

Stafford, Tess M. 2015. "Indoor air quality and academic performance." *Journal of Environmental Economics and Management*, 70: 34–50.

Stansfeld, SA, B Berglund, C Clark, I Lopez-Barrio, P Fischer, E Öhrström, MM Haines, J Head, S Hygge, I van Kamp, and BF Berry. 2005. "Aircraft and road traffic noise and children's cognition and health: a cross-national study." *The Lancet*, 365(9475): 1942–1949.

United States Department of Transportation. 2014. "FAA Faces Significant Obstacles in Advancing the Implementation and Use of Performance-Based Navigation Procedures Federal Aviation Administration." Report No. AV-2014-057.

van Kempen, Elise, Irene van Kamp, Erik Lebret, Jan Lammers, Harry Emmen, and Stephen Stansfeld. 2010. "Neurobehavioral effects of transportation noise in primary schoolchildren: a cross-sectional study." *Environmental Health*, 9: 25.

Vienneau, Danielle, Apolline Saucy, Beat Schäffer, Benjamin Flückiger, Louise Tangermann, Massimo Stafoggia, Jean Marc Wunderli, and Martin Röösli. 2022. "Transportation noise exposure and cardiovascular mortality: 15-years of follow-up in a nationwide prospective cohort in Switzerland." *Environment International*, 158: 106974.

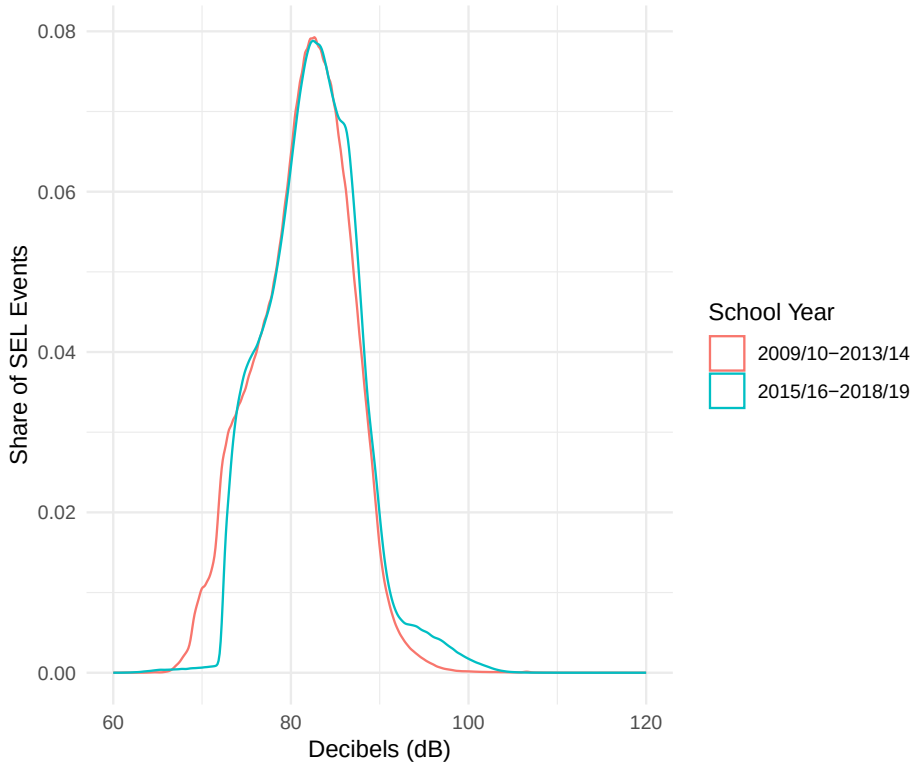
von Graevenitz, Kathrine. 2018. "The amenity cost of road noise." *Journal of Environmental Economics and Management*, 90: 1–22.

Wen, Jeff, and Marshall Burke. 2022. "Lower Test Scores from Wildfire Smoke Exposure." *Nature Sustainability*, 66–95.

WHO. 2011. "Burden of disease from environmental noise: quantification of healthy life years lost in Europe." World Health Organization. Regional Office for Europe.

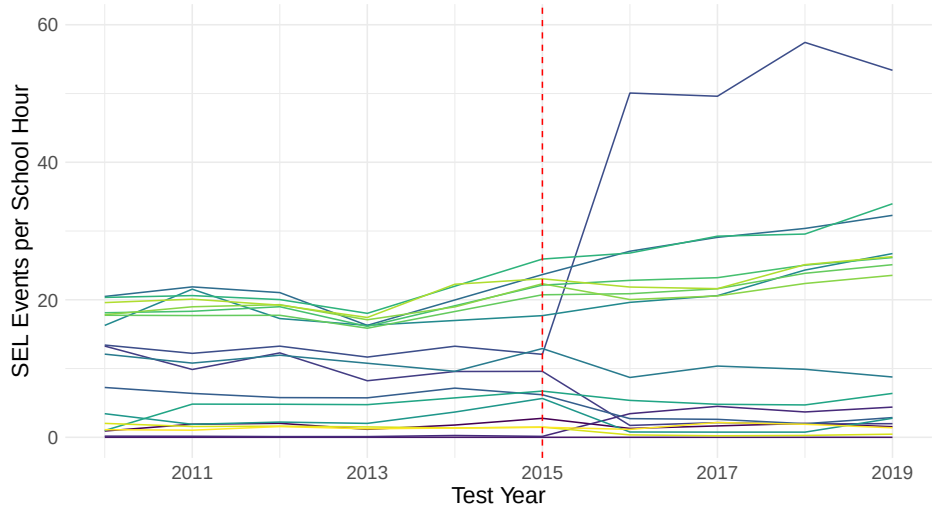
Zhang, Xin, Xi Chen, and Xiaobo Zhang. 2018. "The impact of exposure to air pollution on cognitive performance." *Proceedings of the National Academy of Sciences*, 115(37): 9193–9197.

Appendix Figure 1: Distribution of Sound Exposure Level Events Recorded by The Port of Seattle's Monitor System before and after PBN Implementation



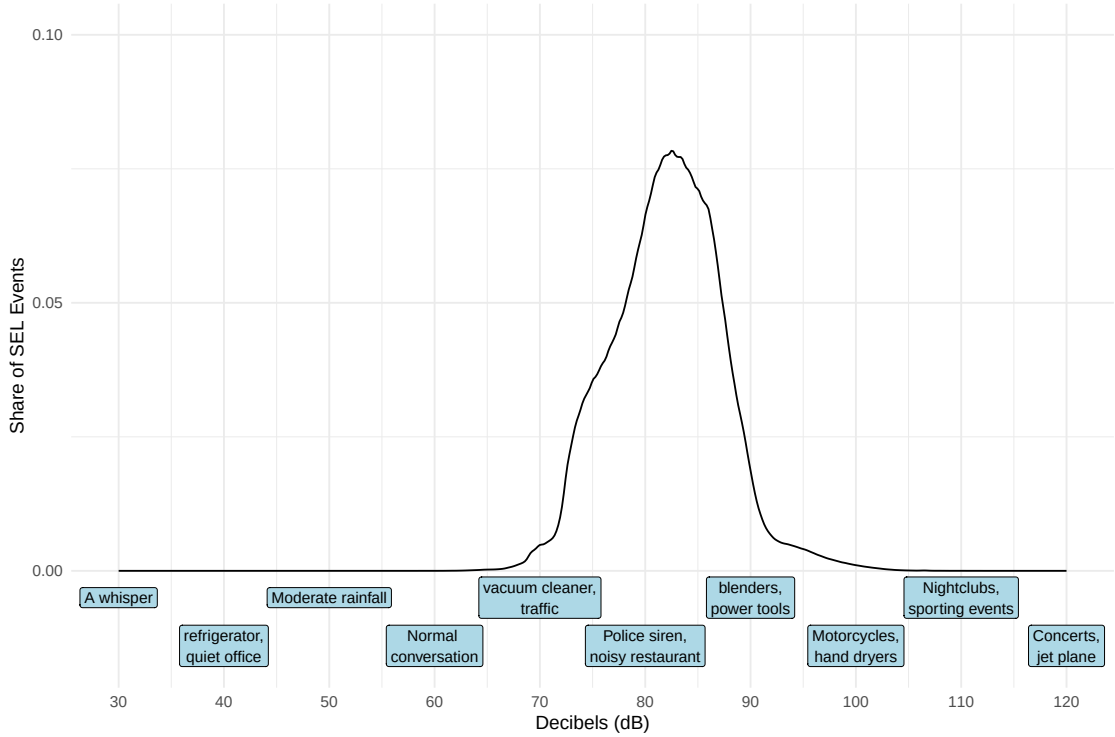
Notes: Monitor sample is limited to those that operated from 2009 through 2019.

Appendix Figure 2: Average Number of Loud Noise Events per School Hour by Noise Monitor and School Year



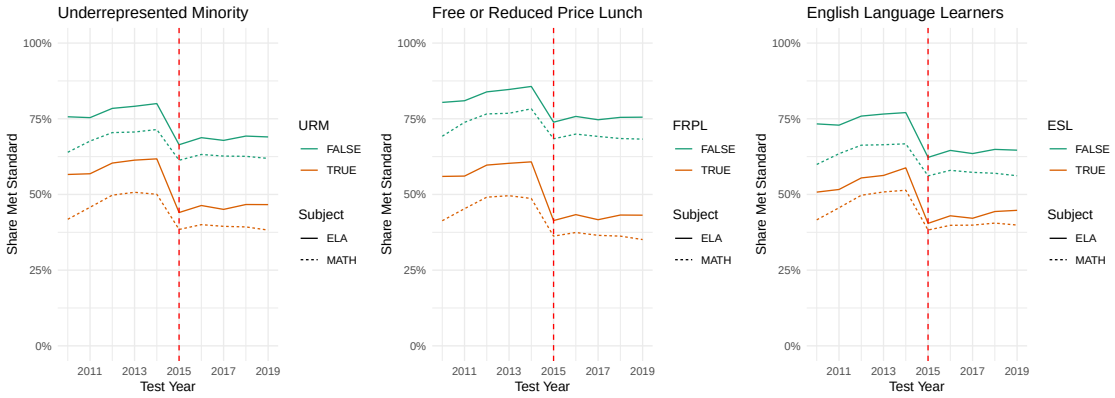
Notes: Each line depicts the annual average number of SEL events per school hour for a noise monitor.

Appendix Figure 3: Distribution of Sound Exposure Level Events Recorded by The Port of Seattle's Monitor System

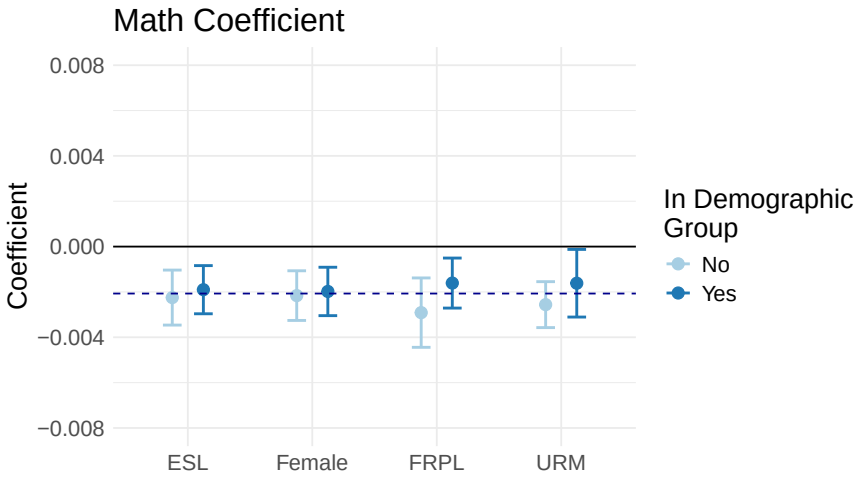


Notes: Monitor sample is limited to those that operated from 2009 through 2019. Density is for all sound exposure level events recorded by these monitors from 2009 through 2019.
Decibel level examples come from MDHearing.

Appendix Figure 4: Impact of Test Regime Change by Demographic Group

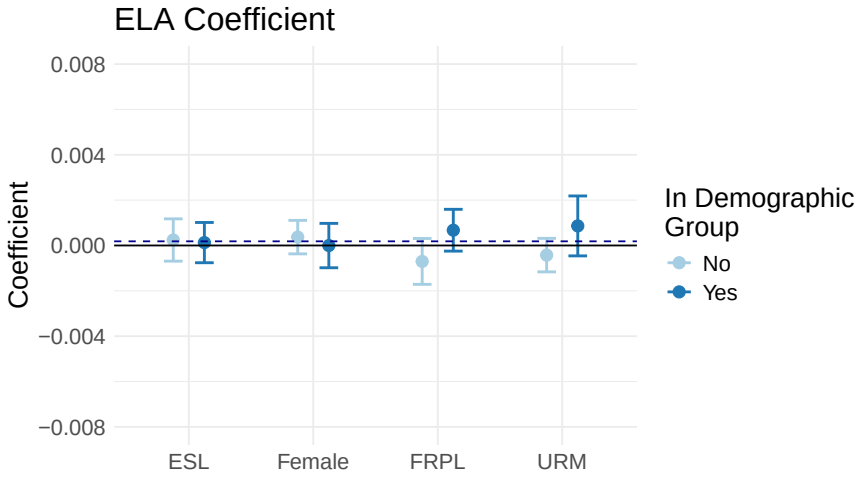


Appendix Figure 5: Heterogeneity of Treatment Effects (Math)

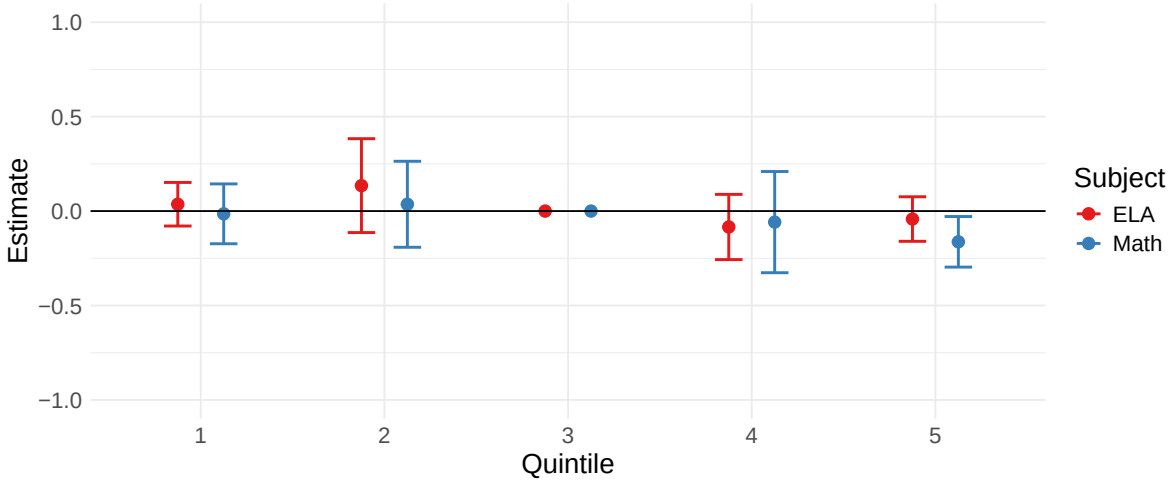


Dashed line represents full sample estimate. "ESL" denotes student's primary language is not English. "FRPL" denotes student receives free or reduced-price lunch. "URM" denotes underrepresented minority student.

Appendix Figure 6: Heterogeneity of Treatment Effects (ELA)



Dashed line represents full sample estimate. "ESL" denotes student's primary language is not English. "FRPL" denotes student receives free or reduced-price lunch. "URM" denotes underrepresented minority student.



Appendix Figure 7: Estimate of Increased Share of Days in Flight Noise Leq Quintiles on Annual Standardized Test Performance